Detecting the anomalous activity of a Ship's Engine

Report on Experiments & Results

By Mahmood Ahmad

UNIVERSITY OF CAMBRIDGE
INSTITUTE OF CONTINUING EDUCATION

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1. Problem Statement

Apply anomaly detection techniques of data science, on a real dataset of a ship's engine function (Mohakul, D. *et al.*, 2023) to detect anomalous engine activity, which can be forwarded to the ship engineer to investigate further.

2. Description of Provided Data set

The dataset comprises of the following six features which are continuously monitored to evaluate the engine's status as 'good' or 'bad':

- `Engine rpm`
- `Lub oil pressure`
- `Fuel pressure`
- `Coolant pressure`
- `lub oil temp`
- `Coolant temp`

All these feature are numerical. The data set consists of 19535 samples. We also checked that there were no null values in all of samples. No repeated rows were found in the data set. Initial statistics were obtained, including the range values beyond the 95% percentile of at least two features.

2.1 Exploratory Data Analysis (EDA)

The boxplots of each feature found outliers in all features (Figure 1). The pair-wise scatter plots showed no correlation between these features (Figure 2), with the only exception of 'Coolant temp' that was horizontal with respect to all other features. This pair-wise heatmap also confirmed this (Figure 3). The individual histograms for every feature is depicted in Figure 4, indicating right skew in the distribution of all features except for 'Coolant temp'.

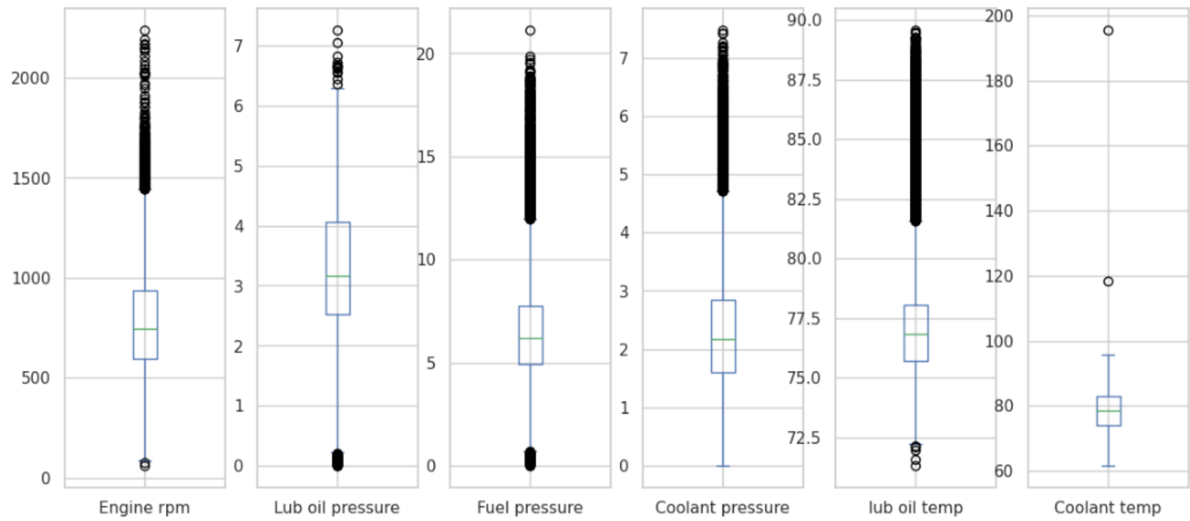

Figure 1: Boxplots of the provided features in Ship's Engine dataset.

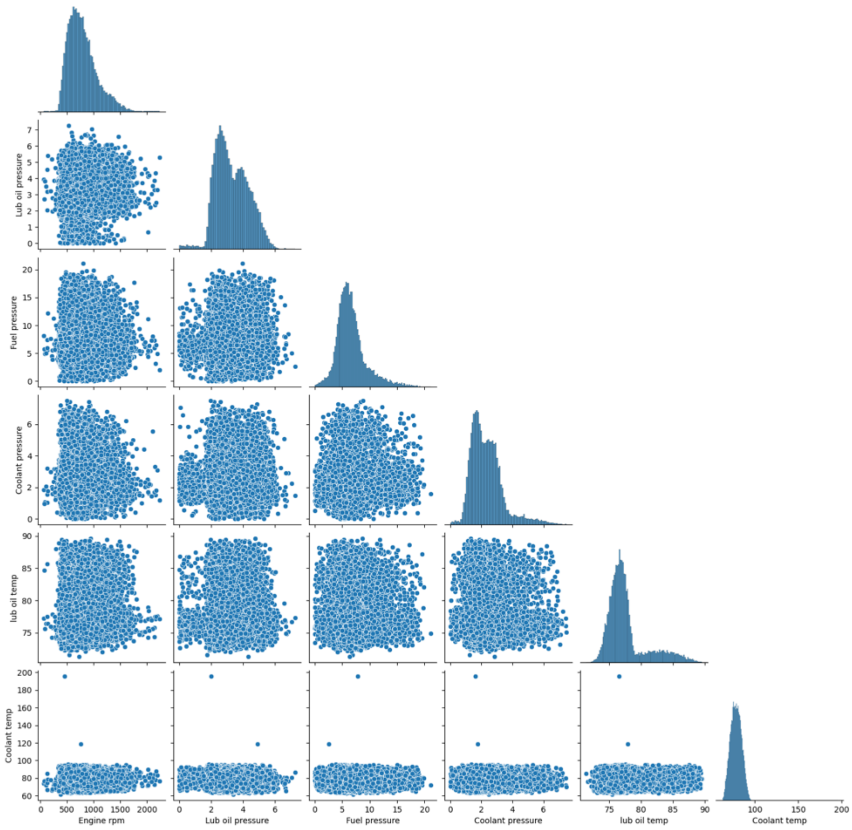

Figure 2: Pair-plots of all features in the provided dataset.

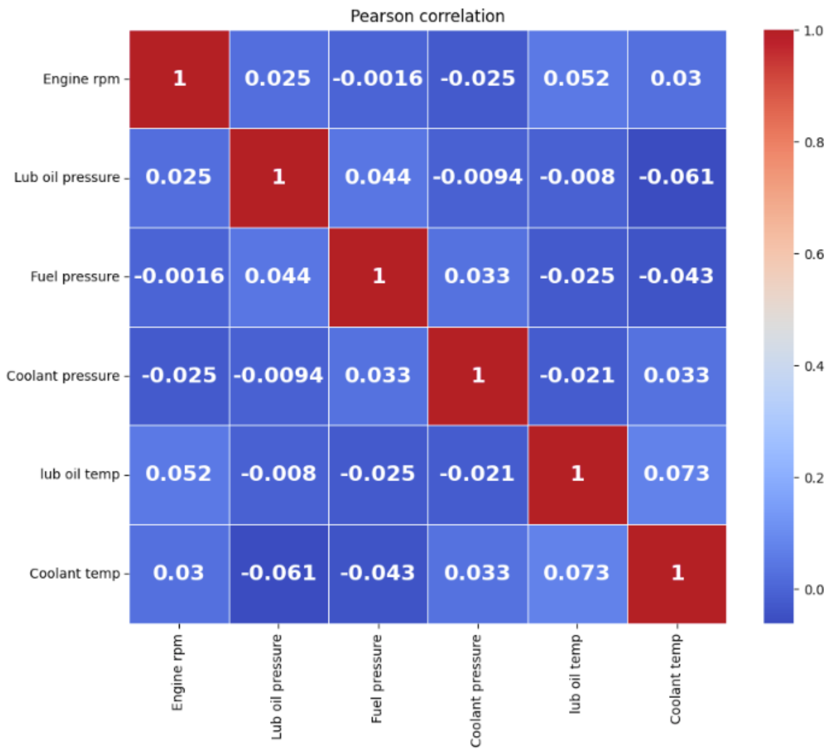

Figure 3: *Pair-wise Pearson's Correlation Coefficient between Ship's Engine features.*

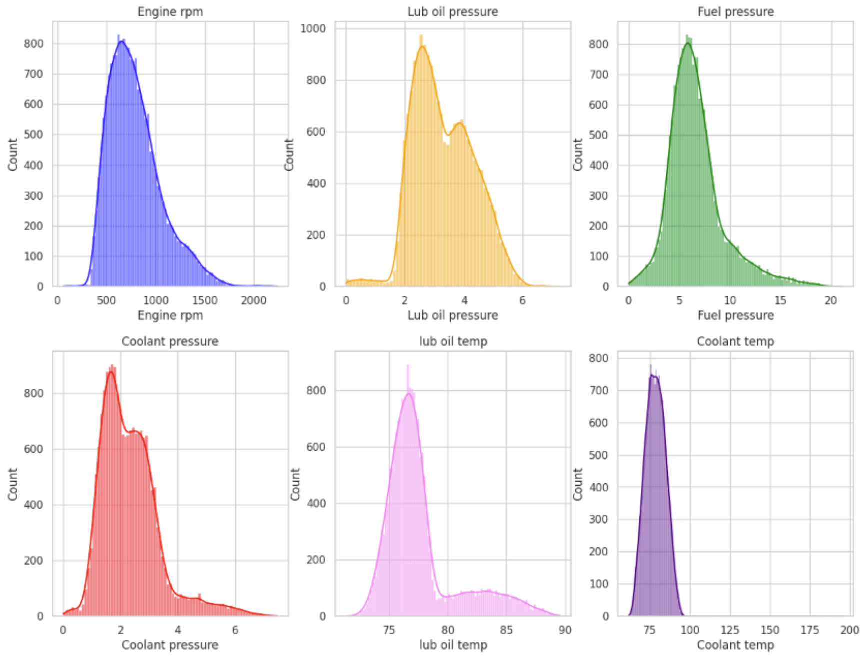

Figure 4: Individual data distributions for every features.

3. Anomaly Detection using Statistical Methods

The Inter-quartile Range (IQR) method was applied for at least one, at least two and at least three features in the same data row.

	Engine rpm	Lub oil pressure	Fuel pressure	Coolant pressure	lub oil temp	Coolant temp	Engine rpm_anomaly	Lub oil pressure_anomaly	Fuel pressure_anomaly	Coolant pressure_anomaly	lub oil temp_anomaly	Coolant temp_anomaly
0	682	2.391656	4.617196	2.848982	76.272417	69.884609	0	0	0	0	0	0
1	605	5.466877	6.424361	5.727520	73.222679	74.907314	0	0	0	1	0	0
2	658	3.434232	3.680896	1.678708	88.089916	78.704806	0	0	0	0	1	0
3	749	2.094656	7.120927	1.639670	77.661625	82.386700	0	0	0	0	0	0
4	676	3.538228	5.956472	3.225336	75.226352	67.153220	0	0	0	0	0	0

Figure 5: Dataset with column-wise anomalies as separate columns.

The guideline of 1% to 5% anomalies, means the range of 196 to 976 observations out of a total 19535 samples (rows). For at least one outlier, there were 4636 samples; 422 samples had at least two or more outliers, and only 11 samples had three or more outliers. No samples had four or more outliers. The number 422 was found within range (Figure 6).

	Index	Engine rpm_anomaly	Lub oil pressure_anomaly	Fuel pressure_anomaly	Coolant pressure_anomaly	lub oil temp_anomaly	Coolant temp_anomaly
0	113	1	0	0	0	1	0
0	122	1	0	0	0	1	0
0	131	0	0	1	0	1	0
0	144	1	0	0	0	1	0
0	148	1	0	0	0	1	0
...	...	...	...	...	...	...	...
0	19236	0	0	1	0	1	0
0	19306	0	0	0	1	1	0
0	19328	0	0	0	1	1	0
0	19343	1	0	0	0	1	0
0	19433	0	0	1	1	0	0

422 rows × 7 columns

Figure 6: Two or more outliers in features found within the same sample.

4. Anomaly Detection using Unsupervised Machine Learning (ML) Techniques

Two unsupervised ML techniques were used for this project:

4.1 One-Class Support Vector Machine (OCSVM) Method

This method requires the data set to be scaled (Géron, 2022). We used the OCSVM model for the scaled features' data, with the radial basis function RBF as the kernel, and adjusted the two parameters of gamma and nu (results in Table 1) to obtain a number of anomalies within the required range.

gamma	nu	Number of Anomalies	Range: 1% - 5%
0.5	0.05	1043	Out of range
0.5	0.01	611	Within range
0.5	0.045	965	Within range
0.1	0.045	880	Within range
0.1	0.1	1951	Out of range
0.5	0.1	1984	Out of range
0.6	0.05	1127	Out of range
0.7	0.05	1234	Out of range

Table 1: Adjusting OCSVM parameters gamma and nu, and results for the given dataset.

Figure 7 the result when we selected for a near-optimum number of 965 anomalies that lie in the required range. Figure 8 presents a 2D visualisation of best two features selected through the Principal Component Analysis (PCA).

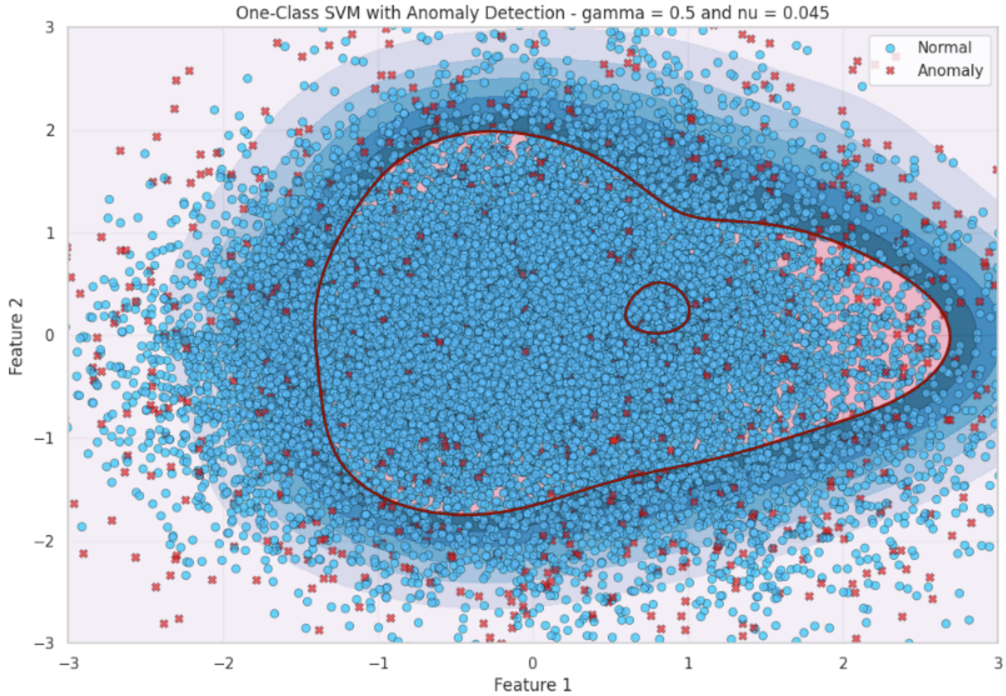

Figure 7: Output of OCSVM for the given dataset with 1947 anomalies.

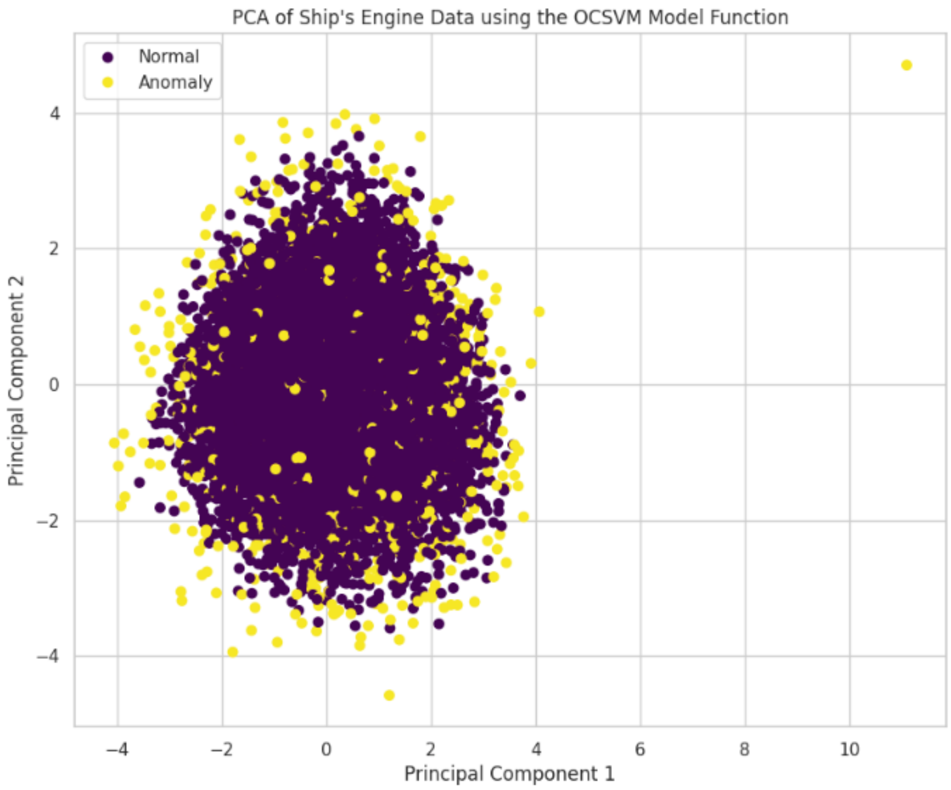

Figure 8: 2D Visualisation using the PCA for the OCSVM method.

4.2 The Isolation Forest (IF) Method

This method is based on decision trees and does not require scaling (Cardona, 2024). Unlike other ML methods, it focuses on isolating the anomalies before separating the normal data samples. Out of its parameters, only the contamination parameter seemed effective (see Table 2).

n_estimators	contamination	Random state	Number of anomalies	Range: 1% - 5%
100	0.1	42	1954	Out of range
100	0.05	42	977	border
100	0.049	42	958	Within range
150	0.1	42	1954	Out of range
150	0.1	84	1954	Out of range
75	0.05	84	977	Border
75	0.049	84	958	Within range

Table 2: Experiments using the Isolation Forest (IF) technique for the given dataset.

The contamination parameter value of 0.049 produced 958 anomalies within the required range. The PCA-based 2D visualisation indicates a considerable number of commonly detected anomalies with the OCSVM method.

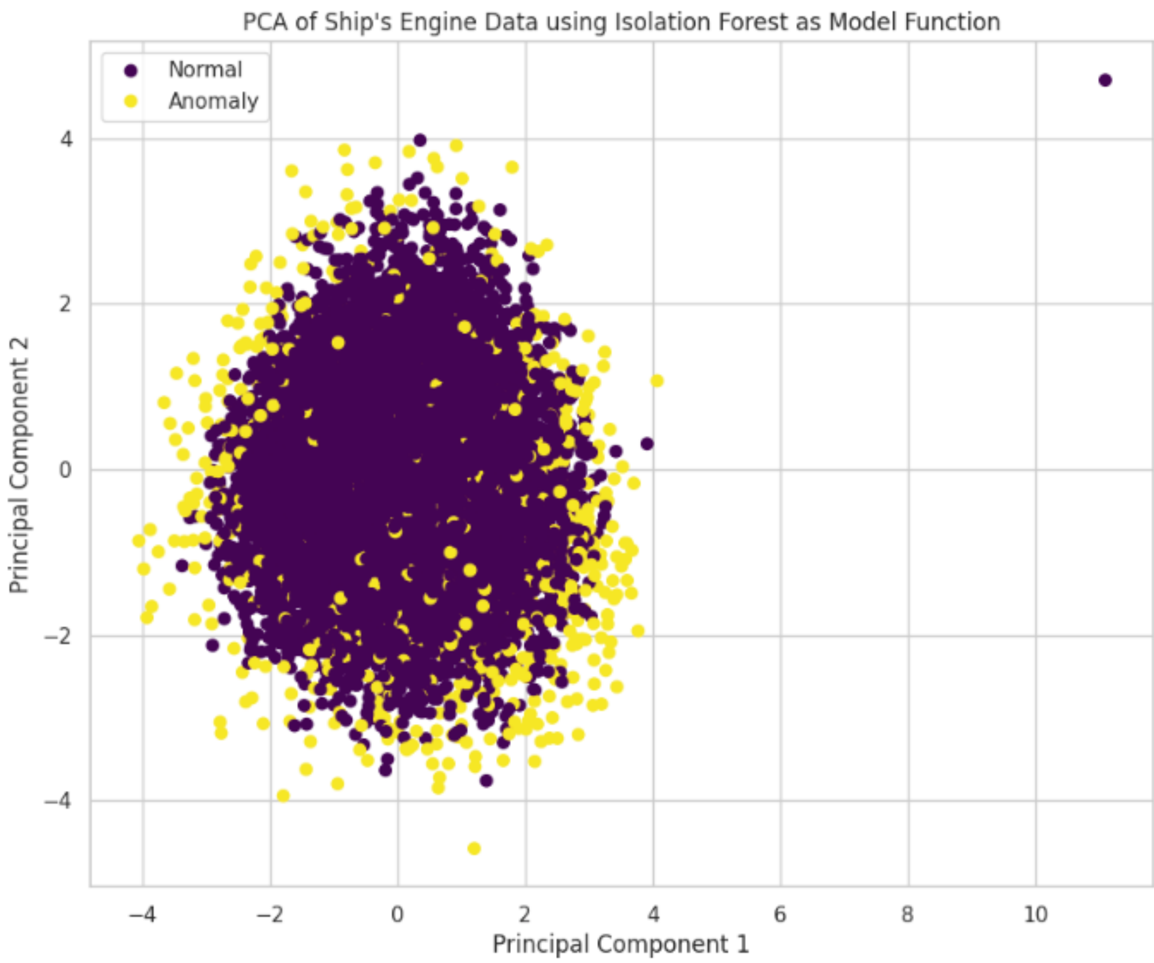

Figure 9: The 2D Visualisation using the PCA for Isolation Forest.

5. Discussion on Results

The IQR method for two anomalies in a sample proved less effective than the ML-based technique. The OCSVM and Isolation Forest methods, although computationally expensive for this large dataset, produced about 5% of samples as anomalies, once their parameters were adjusted.

There were pair-wise shared anomalies among the three methods, other than 190 samples that were commonly detected by all of the three methods. These are detailed in Figure 10.

All of the three methods identified the same 190 samples, which should be the first set of anomalies provided to the ship engineer (Figure 10), followed by the largest set of common 371 anomalies between the OCSVM and IF methods, and so on.

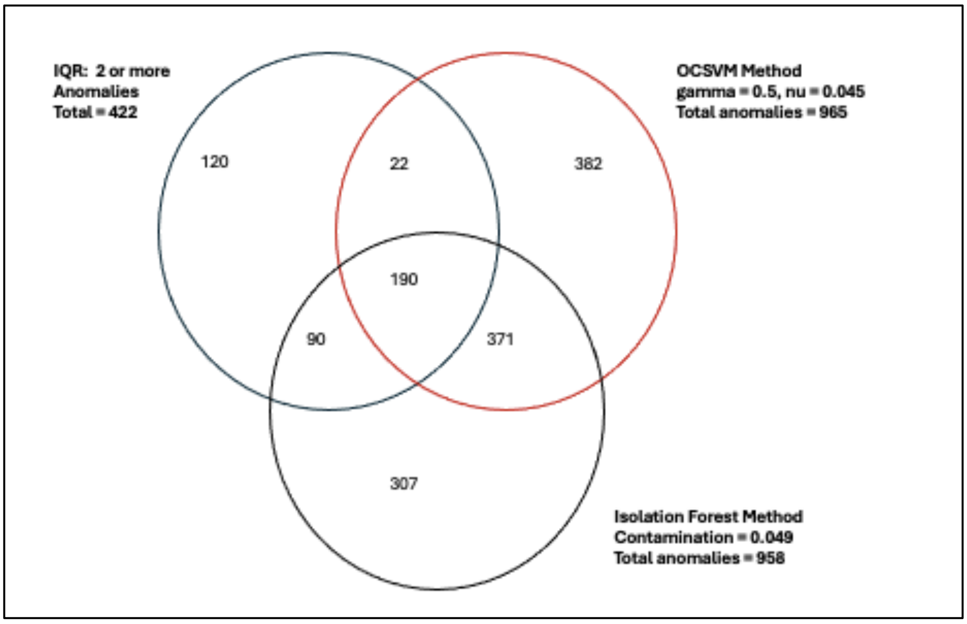

Figure 10: Commonly detected anomalies among the three methods.

6. Conclusions

A dataset like the ship's engine, that does not contain the target variable, challenges a data scientist to extract information from it as much as possible. In this case, the presence of a target variable like 'Engine status', having 'good' or bad' values, would defeat the purpose of anomaly detection in rest of the features.

Single unusual feature value may not indicate any malfunction in an engine. An unusually high 'Engine rpm' value may be due to the ship sailing fast, thus not necessarily an anomaly. However, when two or more features indicate an outlier in IQR method, it would be imperative for us to discuss such samples with the domain experts (ship engineers). For example, if the IQR method indicates an outlier in three features 'Fuel pressure', 'Coolant pressure' and 'lub oil temp' in the same sample, then how likely this is to be an anomaly.

7. References

Mohakul, D. *et al.* (2023) 'CONDITION BASED PREDICTIVE MAINTENANCE ON SHIP'S MAJOR EQUIPMENT USING AI,' *International Journal of Scientific Research in Engineering and Management (IJSREM)*, 7(2). https://doi.org/10.55041/IJSREM17699.

Géron, A. (2022) *Hands-on Machine Learning with Scikit-Learn, Keras & TendorFlow: Concepts, Tools and Techniques to Build Intelligent Systems*. Third Edition. O' Reilly Media, ISBN-10: 1098125975; ISBN-13: 978-1098125975.

Cardona, A. (2024), Applying Statistics and core data science techniques in business: Anomaly Detection, provided on Canvas, Course 1, CAM_DS_101 DS Cohort 2, University of Cambridge Institute of Continuing Education (ICE), URL: https://platform.fourthrev.com/courses/743 .

Pandas (2024), pandas User Guide, URL: https://pandas.pydata.org/docs/user_guide/index.html, accessed on August 25, 2024.